

LUEDENSCHEID, Germany

WMO: 104180

Lat: 51.2453N

Lon: 7.6425E

Elev: 1278

StdP: 14.03

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.2	19.0	8.3	8.8	20.9	12.6	10.9	22.3	24.6	41.6	22.2	42.5	5.2	80	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	81.9	64.6	78.3	63.6	75.0	62.5	67.0	77.3	65.4	74.6	63.8	72.1	6.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	92.3	69.9	62.0	87.2	68.2	60.6	82.9	67.0	32.2	77.9	30.9	74.6	29.7	72.4	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.1	16.6	14.5	DB	9.8	88.2	6.1	2.8	5.4	90.2	1.9	91.8	-1.5	93.4	-5.9	95.4	(4)
(5)			WB	9.2	70.2	6.0	1.5	4.8	71.2	1.3	72.1	-2.1	73.0	-6.5	74.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	33.5	34.8	39.5	46.6	53.7	58.3	62.0	61.8	55.8	48.9	40.9	34.5
	DBStd	12.38	7.73	7.61	7.30	7.50	7.10	6.57	6.18	5.83	5.84	6.66	6.43	7.36
	HDD50	2342	511	427	331	153	43	5	0	0	11	102	278	481
	HDD65	6461	975	847	790	552	355	219	137	136	283	499	722	946
	CDD50	1463	0	1	5	51	158	253	372	365	185	68	5	0
	CDD65	108	0	0	0	0	5	17	43	36	7	0	0	0
	CDH74	917	0	0	0	4	43	166	347	315	41	1	0	0
	CDH80	187	0	0	0	0	4	28	78	75	2	0	0	0
Wind	WSAvg	6.4	7.4	7.8	7.1	6.2	6.0	5.7	5.7	5.3	5.6	6.5	6.5	7.1
Precipitation	PrecAvg	48.1	5.1	4.1	3.7	2.7	3.1	3.6	4.2	4.2	3.7	3.8	4.6	5.4
	PrecMax	64.3	10.2	10.6	8.1	4.7	7.0	6.5	7.2	9.3	8.7	12.2	8.0	12.5
	PrecMin	36.1	0.5	1.1	0.5	0.2	1.2	0.8	1.7	1.5	0.9	1.5	0.2	1.5
	PrecStd	7.1	2.5	2.3	2.1	1.1	1.5	1.3	1.7	2.1	1.9	2.1	2.0	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.2	54.2	62.4	73.7	78.8	84.0	86.4	86.9	78.8	69.1	57.9	51.8
		MCWB	48.1	46.2	50.1	55.6	61.1	65.3	66.0	67.6	63.0	60.0	51.3	48.7
	2%	DB	48.8	50.0	58.0	68.5	73.7	78.7	81.9	81.3	74.1	64.6	54.5	49.5
		MCWB	46.3	45.2	48.0	54.2	59.4	63.7	64.7	64.1	62.7	57.3	50.3	47.3
	5%	DB	46.4	47.5	54.0	63.9	70.3	74.7	77.7	77.4	69.5	61.0	51.9	47.2
		MCWB	44.6	44.3	46.9	52.1	57.4	62.1	63.8	63.8	60.2	55.9	49.4	45.5
	10%	DB	43.5	45.1	50.1	59.7	66.7	70.8	73.9	73.2	65.6	58.2	49.6	44.6
		MCWB	41.7	42.7	45.5	50.1	55.9	60.4	62.7	62.6	58.4	54.5	47.3	42.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.6	48.5	52.6	57.6	64.8	68.1	68.9	69.1	66.1	61.3	53.1	50.2
		MCDB	50.2	50.7	58.7	68.7	74.0	79.0	80.7	82.5	73.7	67.0	55.4	51.3
	2%	WB	46.8	46.5	50.1	55.3	61.8	65.6	66.8	66.7	63.5	58.5	51.1	47.8
		MCDB	47.9	48.9	55.4	65.6	70.8	75.0	76.7	77.6	71.6	62.6	53.4	49.1
	5%	WB	44.4	44.7	48.0	53.4	59.3	63.4	65.2	64.9	61.5	56.6	49.6	45.5
		MCDB	46.1	46.8	52.2	61.7	67.2	71.6	74.4	73.8	68.1	60.3	51.9	47.0
	10%	WB	41.9	42.7	45.9	51.2	56.8	61.5	63.5	63.2	59.3	54.7	47.4	43.1
		MCDB	43.4	44.8	49.7	58.6	64.4	68.7	71.7	70.8	64.3	57.8	49.3	44.5
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.6	11.7	14.7	15.3	15.7	14.9	15.0	13.2	11.2	7.7	7.3
		MCDBR	9.2	11.8	19.1	21.9	21.6	22.3	21.9	22.2	19.4	14.6	10.3	8.8
		MCWBR	8.0	8.1	11.0	11.2	10.7	10.4	9.1	9.6	9.9	8.8	7.6	7.5
	5% WB	MCDBR	8.8	9.0	14.9	18.8	18.9	19.3	19.5	18.7	17.1	12.7	9.1	8.0
		MCWBR	8.2	7.3	9.5	10.2	10.4	10.2	9.1	9.5	9.3	8.5	7.4	7.7
Clear-Sky Solar Irradiance	taub	0.307	0.331	0.361	0.394	0.399	0.400	0.408	0.402	0.373	0.348	0.328	0.304	
	taud	2.419	2.378	2.319	2.265	2.281	2.294	2.270	2.312	2.383	2.442	2.428	2.402	
	Ebn at Noon	227	252	264	267	271	272	267	262	257	242	216	207	
	Edh at Noon	19	26	33	39	40	40	40	36	30	24	19	17	
All-Sky Solar Radiation	RadAvg	233	447	802	1288	1547	1676	1593	1359	993	584	272	177	
	RadStd	35	83	120	148	145	151	200	128	122	56	36	23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page